Dynamic Team

Business Entity Resolution | Amazon ML Challenge 2026 | IIIT Bangalore

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Problem Understanding

Three sources, one question per record: which Source 1 entity is this, if any?

HOW WE FRAMED IT

1. **Assignment, not pair classification**
   Each Source 2 or 3 record belongs to at most one Source 1 entity. So we rank its candidates, keep the best one and decide: accept it, or leave the record unmatched.

2. **Precision first, entity by entity**
   Macro $F_{0.5}$ is averaged over entities. One wrong merge costs about 2.7 times what one correct match earns, and an entity with no true match scores either 1 or 0.

3. **Read the data before building the model**
   Distractor records are built from real entities, and the test set is not distributed like the training set. These two findings shaped every stage.

KEY CHALLENGES ACROSS THE THREE SOURCES

18% of Indian names are in native scripts

Noisy copies
Legal forms, typos, look-alike digits, domain names, partial or missing addresses.

26% of records match nothing; about 40% in test

Look-alike distractors
Built from real entities: same street with the number shifted, or one name word swapped.

50% of entities share a name, legal form aside

Namesakes
Without an address (3% of records) such records cannot be told apart.

15% of test entities are in France, with no labels

An unseen country
French names and addresses follow other conventions than the ones we trained on.

12.5 M
training records with 7.64 M labelled pairs

11.7 M
test records with 1.73 M entities to resolve

3.5
true matches per entity; 5.6% of entities have none

The distractors are siblings

Our first models stalled at 0.975. Reading the wrong merges showed why: similarity cannot tell a typo from a sibling.

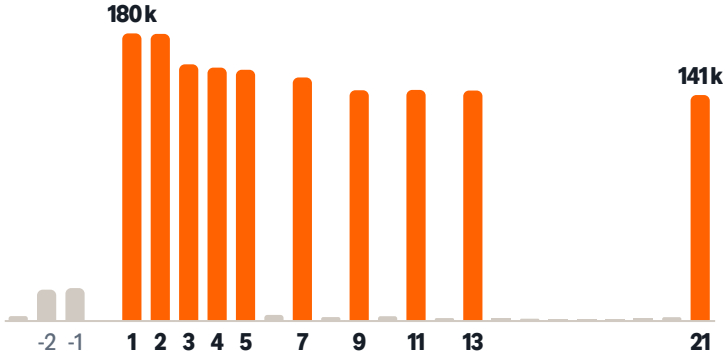
ONE ENTITY FROM THE TRAINING DATA

			similarity	truth
S1	**Peria Sanford Athene** 8513 General Way, Manassas Park City, VA			
S2	**Peria 5anford Athene** MANASSAS PARK CITY, VA, 8513 GENERAL WAY	look-alike digit, reordered	86	match
S2	**PERIA SANFORD ATHENE LLC** 8513 GENRAL WAY, MANASSAS PARK CITY, VA	legal form, typo	96	match
S3	**periasanfordathene.com** 513 General Way, Manassas Park City, Virginia	domain name, digit lost	80	match
S2	**Porter Sanford Athene Corp** 8516 GENERAL WAY, MANASSAS, VA	word swapped, number +3	86	sibling

Similarity: token-sort ratio of name plus address (rapidfuzz), 0 to 100.

HOUSE-NUMBER DIFFERENCE TO THE BEST CANDIDATE

1.65 M records; the number differs by 25 at most and keeps its length



record's house number minus the entity's (bars: number of records)

94.5% sit on ten fixed offsets

2.9% of those are true matches; 69% for other differences

So we stopped scoring how similar two records are and started describing **how they differ**: the signed number difference, which word was swapped, which letter changed. Of the distractors that resemble some entity, 74% are number siblings and 16% word siblings.

Cheap models first, expensive ones last

From 17 trillion possible pairs to 5.85 M matches. Each step passes on only what it could not settle.

1 BLOCK

266 M

pairs, 27 per record

Sparse TF-IDF search inside each state

CPU, 24 min

99.17% of true pairs still present

2 PRUNE

9.73 M

pairs, 5.6 per entity

Small LightGBM on 26 cheap features keeps 2 per record

CPU, 55 min

98.35% still present: the set we submitted

3 MATCH

79%

of records settled here

LightGBM on 66 similarity features, then on what exactly differs

CPU, 2 h

the other **21%** of records go on to step 4

4 CORRECT

2.39 M

pairs still open

Collective evidence plus 4 fine-tuned cross-encoders

GPU, 1.5 h

0.9% of the pairs retrieved in step 1

5 DECIDE

5.85 M

matches, 3.4 per entity

Best candidate per record: $p \geq 0.75$ and 0.1 ahead of the runner-up

3 rules on top

97.50% matched, at **99.93%** precision

Pair counts are for the test set; recall and precision are out-of-fold on the training set.

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Blocking & Candidate Generation

17 trillion possible pairs cut to 5.6 candidates per entity, while 98.35% of the true pairs survive

1 **Search only inside the record's state**

Each record meets only the Source 1 entities of its state, an alternative state (Indiana, PA) and neighbouring states learned from the labels.

Why: there are no postal codes, so the state is the one location key that every source has.

2 **Sparse TF-IDF over several kinds of key**

Name words, consonant skeletons, prefixes, initials, address words, house numbers. We merge three rankings: combined, name only, address only.

Why: exact top-k on CPU, and a record with no address or a changed name is still found by another ranking.

3 **Prune with a small model**

A LightGBM (300 trees, 26 cheap features) keeps the 2 best candidates per record. Exactly this set is scored later and submitted.

Why: keeping 8 lifts the recall ceiling by 0.5 points but changed only 453 of 7.29 M correct final matches.

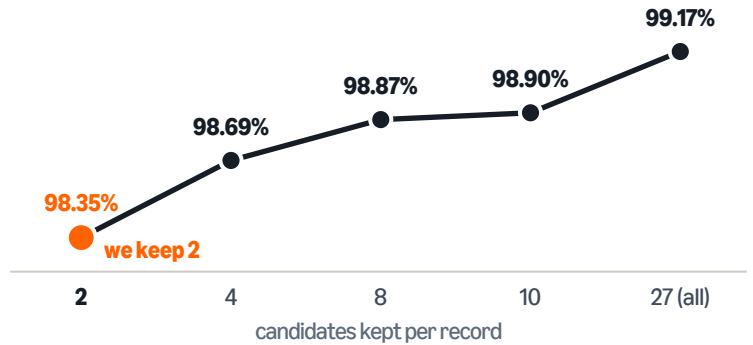
5.6

candidate pairs per entity
(0.98 per record)

98.35%

of true pairs kept
(99.17% before pruning)

RECALL CEILING BY CANDIDATES KEPT PER RECORD



With an address present, retrieval recall is 99.96% (US) and 99.57% (India). The misses are name-only records with very common names.

Matching Model & Features

Four LightGBM stages. Each adds evidence the stage before could not see. Out-of-fold on 2.21 M entities.

STAGE	WHAT IT ADDS	MACRO F0.5	RECALL	WRONG MERGES
STAGE 2 **Matcher**	66 features: string similarity of names and addresses, how two house numbers relate (equal, truncated, same length but different), how many records claim the same entity.	**0.9836**	96.0%	**27.7 k**
STAGE 3 **Differences**	What exactly differs: the signed house-number difference, the word that was swapped, the letter that changed (c for k points to a sibling, g for 6 to a scanning error).	**0.9878**	96.9%	**18.5 k**
STAGE 4 **Collective**	A business keeps its house number within a source: other copies share an unusual number 84% of the time, distractors 4%. Confident copies confirm or reject the uncertain ones.	**0.9894**	97.3%	**14.6 k**
STAGE 4 **+ text models**	Four cross-encoders (33 M to 560 M parameters, MIT licence), fine-tuned on the 0.9 M uncertain pairs. Their scores enter stage 4 as features.	**0.9915**	97.5%	**5.5 k** **80% fewer**

LOG LOSS ON THE 0.9 M UNCERTAIN PAIRS (LOWER IS BETTER)



Why both? Trees read numbers, counts and groups; transformers read the words. Each alone reaches 0.22 to 0.24, together 0.15: they fail on different pairs.

Why LightGBM as the backbone? It scores 540 M pairs in 53 minutes on a laptop, and its features stay inspectable, which is how we found the siblings.

Handling Edge Cases

The three cases named in the brief: what we did about each, and what it bought

SINGLETONS

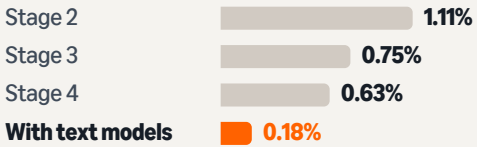
99.8%

of entities with no true match stay empty (98.9% after stage 2)

What we do. One decision per record: only its best candidate, accepted at $p \geq 0.75$ and at least 0.1 ahead of the runner-up. An entity stays empty unless a record clears that bar.

Why. An entity with no true match scores 1 or 0, nothing in between, and 5.6% of entities are like this.

EMPTY ENTITIES WRONGLY GIVEN A MATCH



UNSEEN COUNTRY: FRANCE

15%

of test entities, with no labels in training

Spotted without labels. France had fewer records per entity than the US or India, yet more predicted matches.

Cause. Our model had learned the swapped word by its position: first word in US and Indian names, second in French ones.

Fix. A position-free word-swap veto dropped 31,199 French matches. Leaderboard: 0.9761 to 0.9809.

Then. Transformers trained on US and Indian pairs accepted French siblings, so in France a match needs two votes: +0.0004.

NOISY NAMES AND ADDRESSES

1,347

transliteration words learned from matched training pairs, no outside resource

श्री इंफोटेक: transliterated to shree infotech

OncOlogy, 5olutions: look-alike digits reversed

Pvt Ltd, Private Limited: one legal form, kept in its own field

CAA: an initials key finds Coaching Air Amis

305-307, 305 1/2, 305B: one house number

118 vs 1189: a lost digit is a typo, 80 to 91% true

6106 vs 6101: same length is a sibling, 13% true

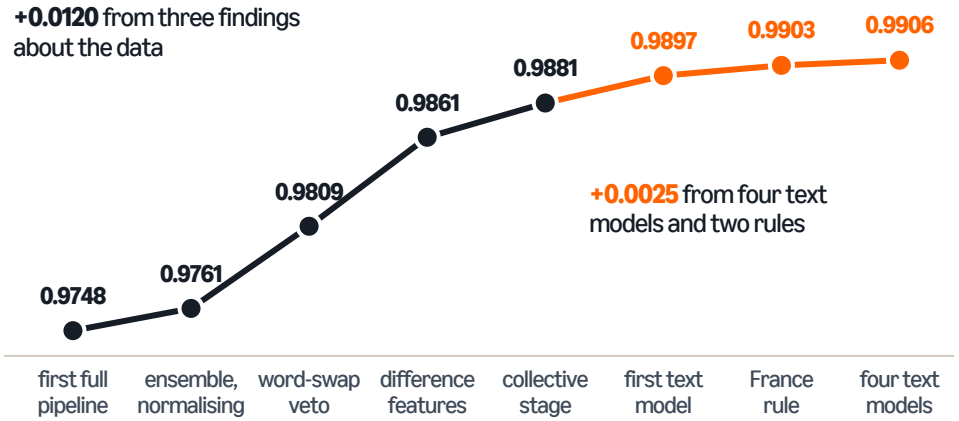
No address (3%): we abstain on shared names; recall 56% against 99.4% with an address

Results & Evaluation

Scored out-of-fold on all 2.21 M training entities with the exact metric, then confirmed on the leaderboard

0.9915	0.9906	99.93%	97.50%	99.8%
macro F0.5, out-of-fold	public leaderboard	pair precision	pair recall	empty entities kept empty

PUBLIC LEADERBOARD, BY WHAT WE CHANGED



HOW WE VALIDATED

- Two folds split by the true entity: no business is in both.
- Every model cross-fitted; exact macro $F_{0.5}$ on all entities.
- Thresholds tuned on a simulated test mix (40% distractors).
- First text model: **+0.00153** predicted, **+0.00152** measured.

PRECISION AGAINST RECALL UNDER F0.5

Entity with 3 true matches, 2 found: one more correct match adds 0.09, one wrong match costs 0.24.

So a match pays off only above **p = 0.73**. Our tuned rule: **p ≥ 0.75** and 0.1 ahead of the runner-up.

Expected-$F_{0.5}$ decoding won on the training mix and lost on the test-like mix; we kept the threshold.

Scalability & Efficiency

Every record gets a fixed number of candidates, so cost grows linearly with the number of records

5.6

candidate pairs per entity

0.98 per Source 2/3 record, 1.8 million times fewer than all possible pairs

MEASURED ON A 10-CORE LAPTOP, 16 GB MEMORY

Normalise	24.2 M records	3 min	**138 k records / s**
Block	20.3 M records	24 min	**14 k records / s**
Stage 1 scoring	540 M pairs	53 min	**170 k pairs / s**

All 15 steps, training included: about 4 h on the laptop plus 1.5 h on a GPU.

The four transformers score 0.9% of the retrieved pairs.

FROM 24 MILLION RECORDS TO BILLIONS

1. **Blocking shards by state.**
   Partitions are independent, so they run in parallel. A hot partition splits again by city, and the matrix becomes an inverted index.

2. **Scoring is linear and stateless.**
   About 27 retrieved and at most 2 kept per record. At our measured speed, stage 1 needs about 440 core-hours per billion records.

3. **Collective features are group-bys.**
   One shuffle on the entity id in batch, a key lookup when records arrive one at a time.

4. **GPU cost stays bounded.**
   Transformers see only the 21% of records still open, and the smallest model alone gave three quarters of their gain: distil to one.

Honest limit. Our index is an in-memory sparse matrix per country (largest: 1.3 M entities). At billions it must become a sharded inverted index. We designed for that but did not build it.

Learnings & Reflections

What worked, what did not, and what we would do differently

WHAT WORKED

Reading errors before tuning models.
Three findings about the data moved the leaderboard by 0.0120. Four transformers and two rules moved it by 0.0025.

Validation we could trust.
Entity-level folds, the exact metric and a test-like class mix. The offline gain matched the leaderboard gain to the fourth decimal.

A cascade.
Cheap models everywhere, expensive ones on under 1% of pairs. All of it fits a laptop plus 1.5 GPU-hours.

WHAT DID NOT

Expected-F0.5 decoding.
Better on the training mix, worse on the test-like mix. We kept a plain threshold.

Transformers in France.
Trained on US and Indian pairs, they accepted French siblings: 0.98924 against 0.98967 without them there.

More of the same.
8 candidates instead of 2, a larger stage 4, a second stage-4 pass, self-training on French pseudo-labels: no gain out-of-fold.

WHAT WE WOULD CHANGE

Two rules tuned on the leaderboard.
The France and sibling-group rules were picked with leaderboard feedback, because France has no labels. We would rather build a French validation set by synthesising siblings.

Name-only namesakes.
65% of our remaining error. A record with no address and a shared name needs more attributes, not a better model.

One model, not four stages.
Distil the cascade and the four transformers into a single matcher that is easier to run and maintain.

Thank you. Questions?

Understand the data first. Spend compute only where the answer is still open.

0.9906
public leaderboard
(0.9915 out-of-fold)

5.6
candidate pairs
per entity

99.93%
pair precision
at 97.5% recall

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Backup: stage by stage

Out-of-fold on all 2.21 M training entities. Test-like mix: the same scores on a simulated mix with 40% distractor records.

DECISION AFTER	MACRO F0.5 TRAINING MIX	MACRO F0.5 TEST-LIKE MIX	PAIR PRECISION	PAIR RECALL	EMPTY ENTITIES KEPT EMPTY	WRONG MERGES
Stage 2: matcher	0.98355	0.98245	99.62%	95.99%	98.89%	27,693
+ word-swap veto	0.98385	0.98289	99.66%	95.99%	98.98%	24,912
Stage 3: differences	0.98779	0.98716	99.75%	96.85%	99.25%	18,539
Stage 4: collective	0.98939	0.98900	99.80%	97.27%	99.37%	14,649
Stage 4 + four text models	**0.99154**	**0.99141**	**99.93%**	**97.50%**	**99.82%**	**5,501**

- Precision, recall and wrong merges are for the training mix. Every row uses the same rule: the best candidate per record, with threshold and margin tuned on the test-like mix.
- The 33 M-parameter text model alone lifts stage 4 from 0.98939 to 0.99100; the other three and a wider stage-4 input add 0.00054.
- Of the 5,501 remaining wrong merges, 2,509 are distractor records and 2,992 are copies of a different entity.

Backup: text models

Four public checkpoints, fine-tuned only on the challenge's training pairs

CHECKPOINT	LICENCE	PARAMETERS	EPOCHS	LOG LOSS ON HELD-OUT UNCERTAIN PAIRS
microsoft/MiniLM-L12-H384-uncased	MIT	33 M	3	0.264
FacebookAI/xlm-roberta-base	MIT	278 M	2	0.256
FacebookAI/xlm-roberta-large	MIT	560 M	2	0.241
microsoft/deberta-v3-base	MIT	184 M	2	0.246
Stage 3 on the same pairs (reference)				0.298

HOW THEY ARE TRAINED

As cross-encoders on "name | address" of the record and of the Source 1 entity. Only on the 0.9 M pairs stage 3 was unsure about, with the same two entity folds, so every training score is out-of-fold.

HOW STAGE 4 USES THEM

Three features per model: its probability, its margin over the record's other candidate, and how many other records it gives to the same entity with confidence.

FAIR PLAY

No external data, API or geocoder. All four checkpoints are MIT-licensed and far below the 8 B-parameter limit. Every other learned resource comes from the training labels.

Backup: where the remaining error is

Out-of-fold loss of the final model: 1 - 0.9915 = 0.0085 macro F0.5

LOSS BY KIND OF RECORD

No address, name shared by several entities



0.0055 65%

Record has an address



0.0022 26%

No address, unique name



0.0008 10%

Each bar: how much the score would rise if that kind of record were decided perfectly. The three were measured separately, so they add up only approximately.

A record with no address and a shared name is **ambiguous in the data**, not mis-modelled. More attributes would fix it; a better matcher would not.

WHAT THE NUMBERS SAY

1. **Recall depends on the address.**
   99.4% of true pairs are found when the record has an address, 55.8% when it has none.

2. **The model knows when it cannot know.**
   Name-only pairs with one namesake: 97% are true, the model says 96%. With five or more namesakes: 16% true, model 18%.

3. **Wrong merges are rare and of two kinds.**
   5,501 of 7.45 M accepted pairs: 2,509 distractor records and 2,992 copies of a different entity.

4. **A third of the misses were never retrieved.**
   Mostly name-only records with very common names.

Backup: alternatives we measured

Each choice was made on the out-of-fold score

CANDIDATES KEPT PER RECORD

KEPT	RECALL CEILING OF TRUE PAIRS
2 (submitted)	**98.35%**
4	98.69%
8	98.87%
10	98.90%
27 (all retrieved)	99.17%

Keeping 8 changed only 453 of 7.29 M correct final matches.

DECISION RULE ON STAGE-2 SCORES

RULE	TRAINING MIX	TEST-LIKE MIX
Threshold 0.75, margin 0.1	**0.98355**	**0.98245**
Threshold 0.70, margin 0.1	0.98371	0.98243
Expected F0.5 per entity	0.98389	0.98211

Expected-F decoding wins when the class mix matches training and loses when distractors increase, so we chose the rule that is best on the test-like mix.

TRIED AND DROPPED: NO GAIN OUT-OF-FOLD OR ON THE LEADERBOARD

- A larger stage-4 model
- A second pass of stage 4 on its own output
- Averaging several stage-4 versions
- Self-training on French pseudo-labels
- Recovering unretrieved records through shared addresses
- Text models deciding France: 0.98924, against 0.98967 without them

Backup: reproducibility

One command from the raw TSV files to both output files

```
bash src/run_all.sh
```

STEPS AND TIME ON A 10-CORE LAPTOP

Normalise and transliterate	**3 min**
Blocking	**24 min**
Stage 1: prune	**55 min**
Stage 2: matcher	**95 min**
Stage 3: differences	**25 min**
Stage 4: collective, three variants	**37 min**
Text models (GPU)	**1.5 h**
Outputs, rules, validation	**7 min**

WHAT WE CHECKED

1. **The package reproduces the submission.**
   On 1 October we re-ran the final stages from the submitted package and its saved text-model scores: both output files came out byte for byte identical.

2. **Fixed seeds on CPU.**
   Every CPU step gives the same result on every run. GPU fine-tuning can differ slightly between runs, so the exact text scores behind the submission are saved in the package.

3. **Only the provided data.**
   No external database, API or geocoder. Hand-written tables hold only general knowledge: state names, legal-form and street abbreviations.

4. **Pinned environment.**
   Python 3.12 with pinned versions; LightGBM, polars, rapidfuzz, PyTorch.